



Ancora
Counselling Services

SUPPORT • GROW • THRIVE

ANCORA • ACTIVITY 3.3

User Interface Design

Navigation • Integration • ERD Alignment

Compiled after System Architecture, Use Cases
& Sequence Diagrams

Group D • ITC327W • September 2026

SUPPORT • LISTEN • EMPOWER



OVERVIEW

What This Activity Covers

01



Required Screens

Flutter (Student) and ASP.NET (Student, Psychologist, Admin) screen inventory

02



Navigation Maps

How users move between screens for each role

03



Mobile ↔ Web Integration

Flutter + ASP.NET sharing the same Supabase backend

04



Interface ↔ ERD Alignment

Mapping every screen function to database entities and fields

05



Overall Design Alignment

How SRS, Use Cases, UI, ERD and both clients fit together

3. Problem Statement



3.1 Current Situation

The Wellness Centre currently depends on the Outlook calendar and paper-based methods to record appointment requests. The admin reviews and approves each request and assigns a psychologist, while student records and statistics are maintained separately in Excel by each psychologist.



3.2 The Problem

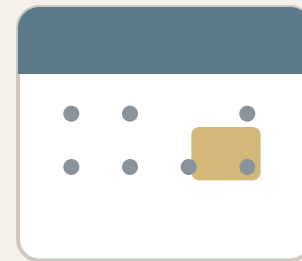
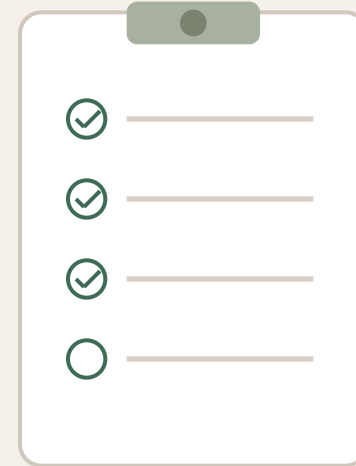
This manual, fragmented process leads to missed appointment requests, double bookings, misplaced student records, and delays in generating accurate statistics. Emergency cases are only prioritised when judged sufficiently serious, and there is no reliable way to search existing student records.

Missed requests

Double bookings

Lost records

Slow reporting



Manual tools fragmented care

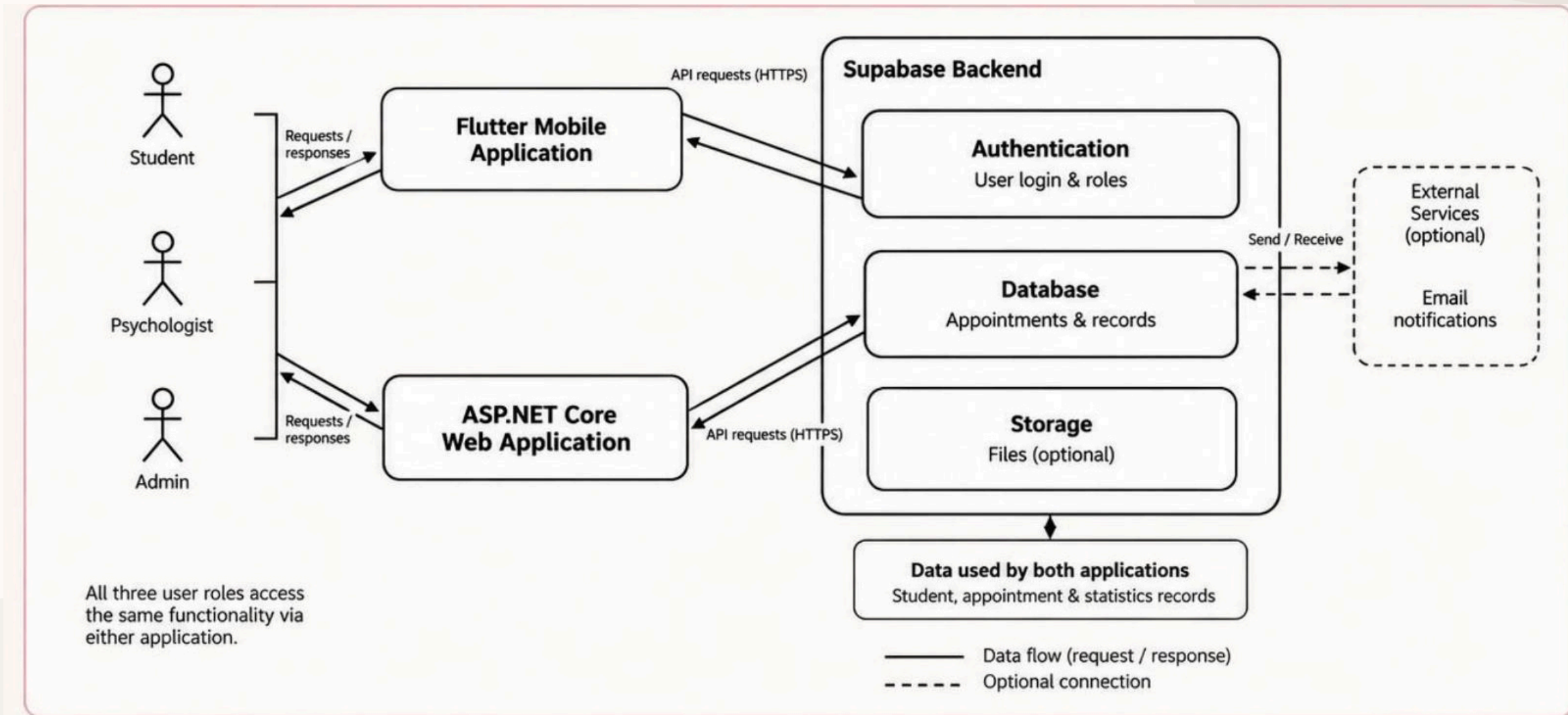
Wellness Centre Stakeholders

CUT Wellness Centre — roles, interests, and contact points that shape Ancora’s requirements.

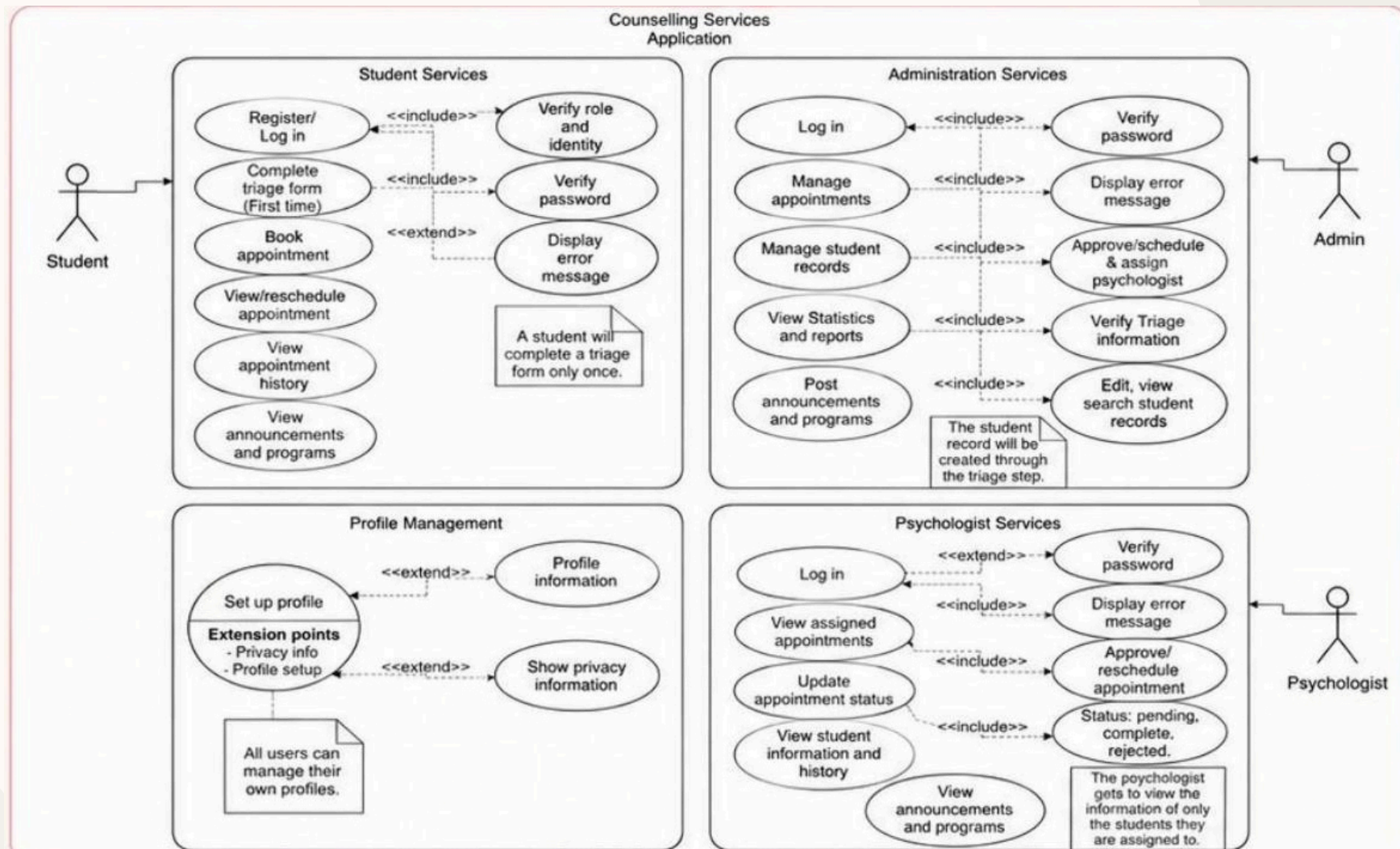
NAME	ROLE	ORGANIZATION	INTEREST / EXPECTATIONS	CONTACT
Gelinda Nkili	Manager, Psychologist	CUT’s Wellness Centre	Won’t use the system every day but will maintain it. Expects a user-friendly, safe and secure platform.	gpillay@cut.ac.za
Ntuthuko Myaka	Psychologist	CUT’s Wellness Centre	Will use the platform as part of his job. Recommends encryption & password protection for confidentiality.	NMyaka@cut.ac.za
Mathabo Mpuru	Administrator	CUT’s Wellness Centre	Responsible for management of student records and appointment bookings.	mmpuru@cut.ac.za

Stakeholder needs were confirmed through structured interviews with the Wellness Centre team.

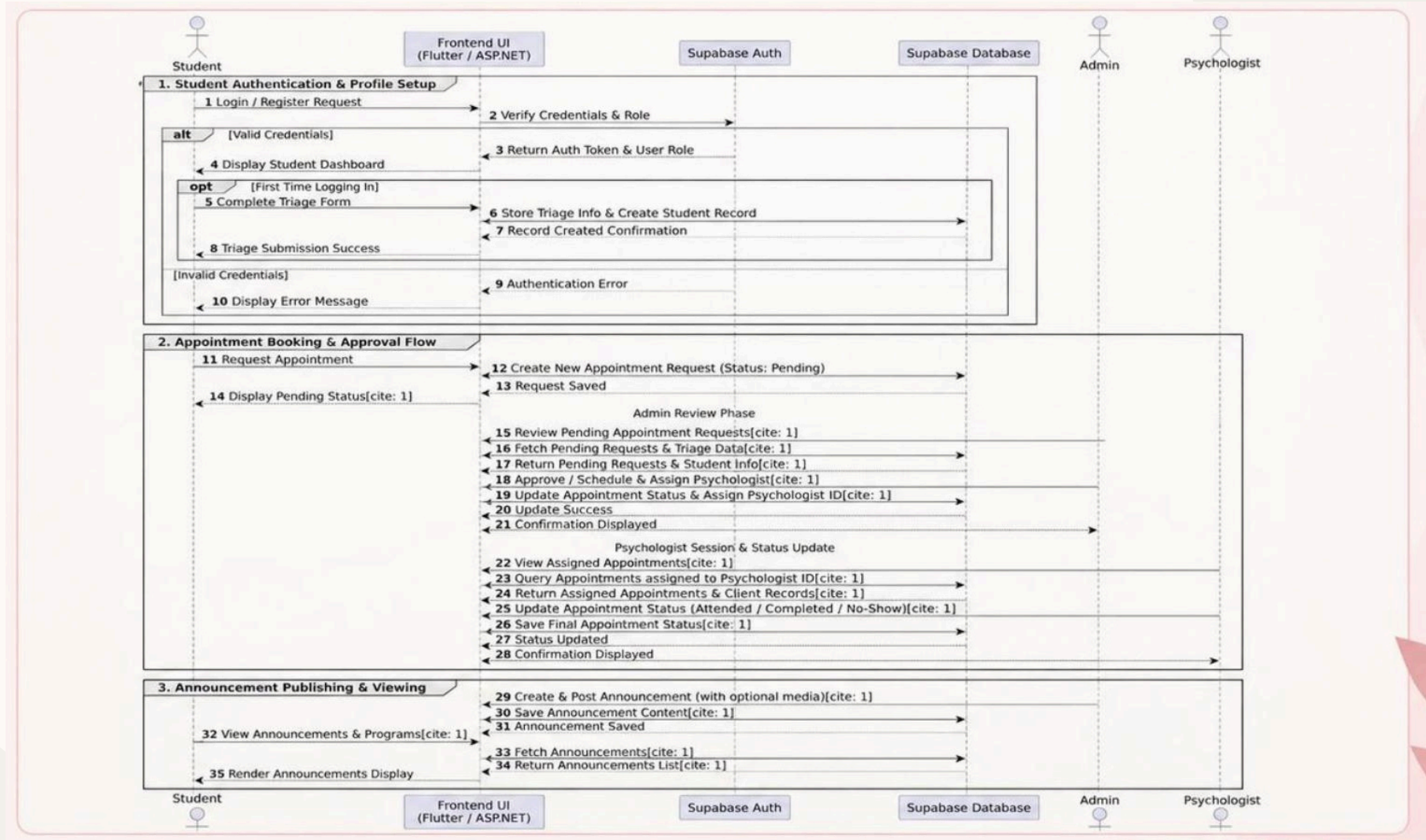
System Architecture Diagram



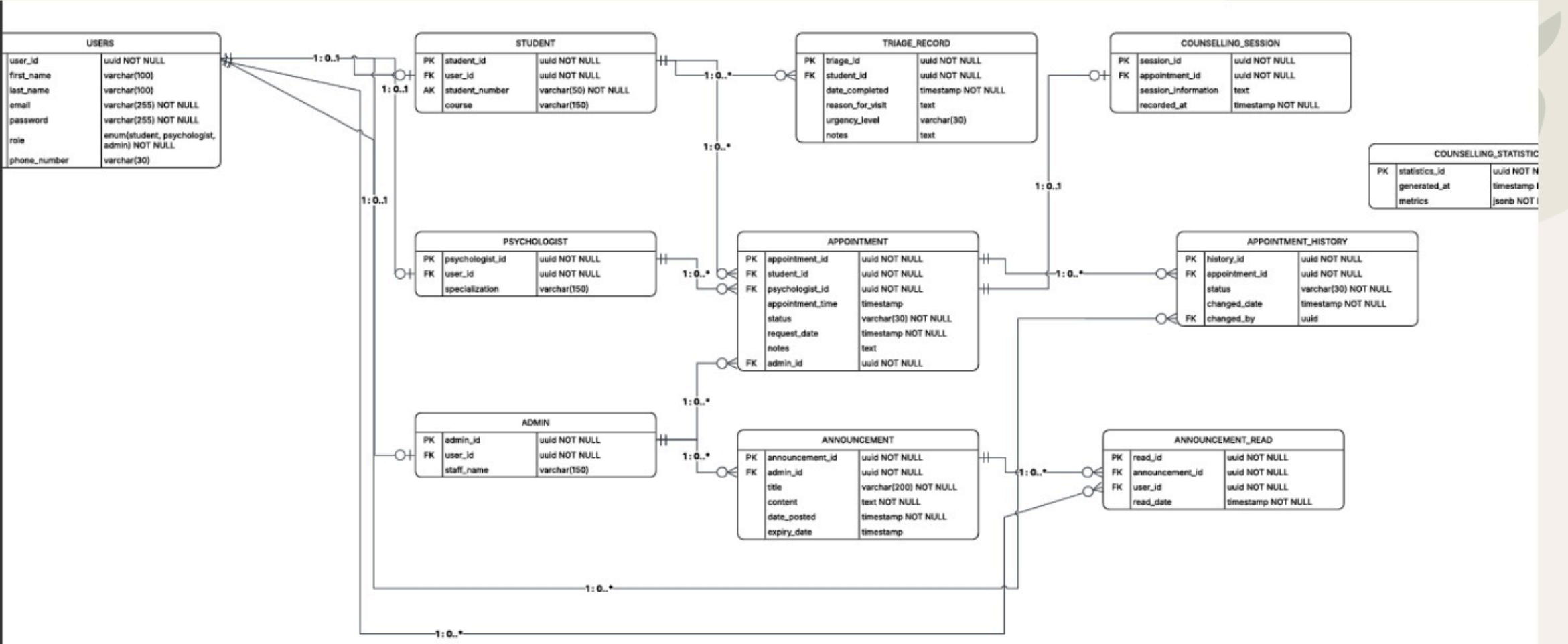
Use Case Diagram



Sequence Diagram



Entity Relationship Diagram



STEP 2 — REQUIRED SCREENS

Flutter Screens (Student)

Flutter Screen	Purpose	Related Use Case
 Login	Securely logs the student into the system	Student Login
 Student Dashboard	Access to main counselling services	View Dashboard
 Request Appointment	Triage information and appointment request	Request Appointment
 My Appointments	View upcoming appointments and status	View / Manage Appointments
 Appointment History	View previous appointments	View Appointment History
 Messages & Announcements	View messages and announcements	View Messages & Announcements
 Profile	View or manage profile information	Manage Profile

STEP 2 — REQUIRED SCREENS

ASP.NET Screens (Student)

ASP.NET provides the same student screens as Flutter. Both apps share the identical feature set and connect to the same Supabase backend.

ASP.NET Screen	Purpose	Related Use Case
 Login	Securely logs the student into the system	Student Login
 Student Dashboard	Access to main counselling services	View Dashboard
 Request Appointment	Triage information and appointment request	Request Appointment
 My Appointments	View upcoming appointments and status	View / Manage Appointments
 Appointment History	View previous appointments	View Appointment History
 Messages & Announcements	View messages and announcements	View Messages & Announcements
 Profile	View or manage profile information	Manage Profile

STEP 2 — REQUIRED SCREENS

ASP.NET Screens (Psychologist & Administrator)

	PSYCHOLOGIST
	Login — Secure login
	Psychologist Dashboard — Today's appointments & pending requests
	Manage Appointments — View, confirm or decline
	Appointment Details — Triage info, notes, status
	Student Records — View relevant student info
	Messages & Announcements
	Profile — View or manage profile

	ADMINISTRATOR
	Login — Secure login
	Administrator Dashboard — Students, appointments, stats
	Manage Students — Search, view, update records
	Manage Appointments — Approve, reject, reschedule
	Appointment Details — Full appointment & triage
	Reports & Statistics — Counselling statistics
	Messages & Announcements — Create & publish
	Settings — User / account settings

Student Navigation Map

From	Action	To
Login	Enter credentials → Log in	Student Dashboard
Dashboard	Request Appointment	Book Appointment
Book Appointment	Triage → Date/Time → Confirm	My Appointments
Dashboard	My Appointments	My Appointments
My Appointments	Select appointment	Appointment Details
Dashboard	Appointment History	Appointment History
Dashboard	Messages & Announcements	Messages & Announcements
Dashboard	Profile	Profile
Any screen	Log Out	Login

STEP 3 — NAVIGATION

Psychologist & Administrator Navigation

 PSYCHOLOGIST	
	Login → Psychologist Dashboard
	Dashboard → Manage Appointments
	Manage Appointments → Appointment Details
	Appointment Details → Student Records
	Dashboard → Student Records
	Dashboard → Messages & Announcements
	Dashboard → Profile
	Any screen → Log Out → Login

 ADMINISTRATOR	
	Login → Administrator Dashboard
	Dashboard → Manage Students
	Dashboard → Manage Appointments
	Manage Appointments → Appointment Details
	Dashboard → Reports & Statistics
	Dashboard → Messages & Announcements
	Dashboard → Settings
	Any screen → Log Out → Login

STEP 6 — INTEGRATION —

Flutter ↔ Supabase ↔ ASP.NET

Both applications use the same Supabase backend so information created or changed in one is immediately available to the other.

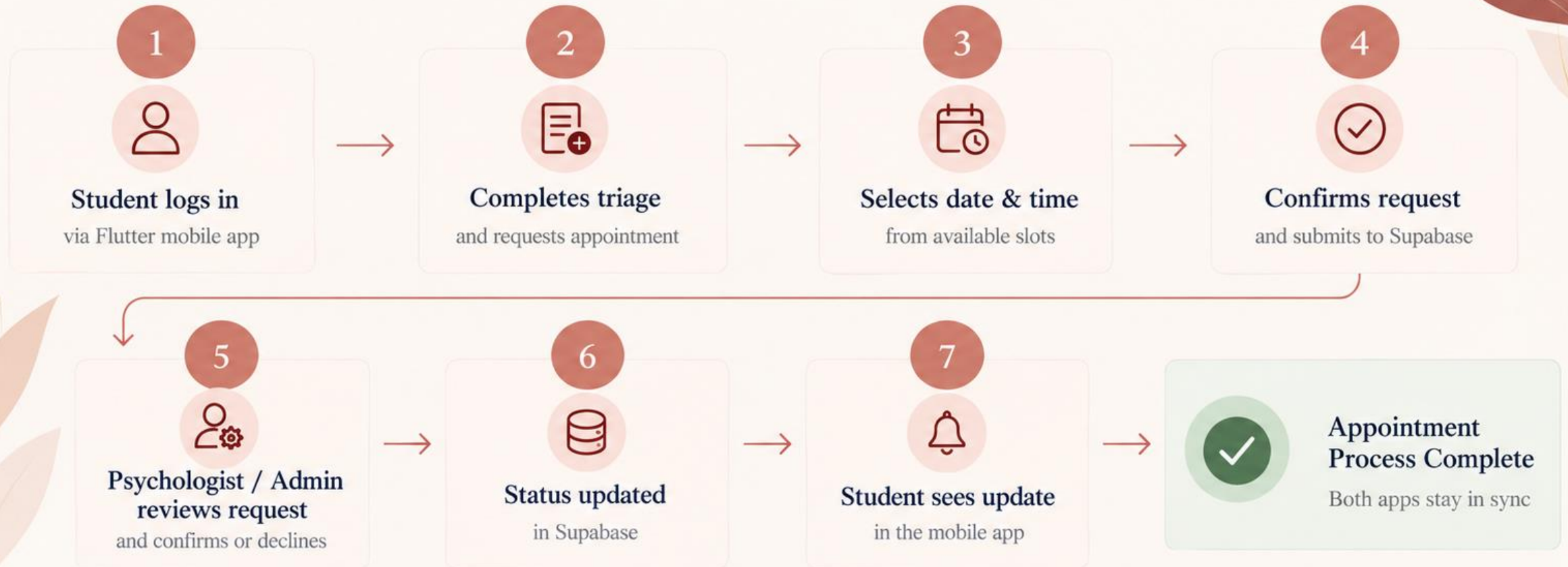


Example data flow

Action (Source)	Supabase (Process)	Result (Target)
 Student submits appointment (Flutter)	 Creates Appointment	 Psych / Admin views request (ASP.NET)
 Psychologist confirms / declines (ASP.NET)	 Updates Status	 Student sees updated status (Flutter)
 Admin creates announcement (ASP.NET)	 Creates Announcement	 Available to all users (Flutter & ASP.NET)

STEP 6 — INTEGRATION

End-to-End Appointment Request Flow



STEP 7 — INTERFACE TO ERD ALIGNMENT

Student Interface ↔ ERD

	Interface	Information / Function	ERD Entity	Attribute / Field
	Login	Email / Password	USER	Email, Password
	Dashboard	Student name	USER	FirstName, LastName
	Request Appointment	Reason, Mood, Info	TRIAGE_RECORD	Reason, Mood, AdditionalInfo
	Request Appointment	Date & Time	APPOINTMENT	AppointmentDate, Time
	My Appointments	Date, Time, Psych, Status	APPOINTMENT	Date, Time, PsychologistID, Status
	Appointment History	Previous appointments	APPOINTMENT	AppointmentID, Status
	Messages	Sender, Recipient, Content	MESSAGE	SenderID, ReceiverID, MessageText
	Announcements	Title & Content	ANNOUNCEMENT	Title, Content
	Profile	Student information	STUDENT / USER	Student details

STEP 7 — INTERFACE TO ERD ALIGNMENT

Psychologist & Administrator ↔ ERD

 PSYCHOLOGIST		
	Login	→ USER (Email)
	Dashboard	→ APPOINTMENT (Date, Status)
	Dashboard	→ STUDENT (StudentID)
	Manage Appointments	→ APPOINTMENT
	Student Records	→ STUDENT / USER
	Appointment Details	→ TRIAGE + APPOINTMENT
	Messages	→ MESSAGE
	Announcements	→ ANNOUNCEMENT

 ADMINISTRATOR		
	Login	→ USER (Email)
	Dashboard	→ STUDENT, APPOINTMENT, ANNOUNCEMENT
	Manage Students	→ STUDENT / USER
	Manage Appointments	→ APPOINTMENT
	Reports & Statistics	→ APPOINTMENT counts
	Messages	→ MESSAGE
	Announcements	→ ANNOUNCEMENT (create)
	Settings	→ USER (account info)

STEP 7 — INTERFACE TO ERD ALIGNMENT

Psychologist & Administrator ↔ ERD

 PSYCHOLOGIST	
 Login → USER (Email)	
 Dashboard → APPOINTMENT (Date, Status)	
 Dashboard → STUDENT (StudentID)	
 Manage Appointments → APPOINTMENT	
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 Appointment Details → TRIAGE + APPOINTMENT	
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 ADMINISTRATOR	
 Login → USER (Email)	
 Dashboard → STUDENT, APPOINTMENT, ANNOUNCEMENT	
 Manage Students → STUDENT / USER	
 Manage Appointments → APPOINTMENT	
 Reports & Statistics → APPOINTMENT counts	
 Messages → MESSAGE	
 Announcements → ANNOUNCEMENT (create)	
 Settings → USER (account info)	

Requirements → Use Cases → Sequence → ERD → UI

Requirement	Use Case	Sequence	ERD	Interface
Student login	Register / Log in	1. Auth & Profile	USERS	Flutter Sign In
Book appointment	Book appointment	2. Booking & Approval	APPOINTMENT; TRIAGE	Book Appointment
Admin approve / assign	Manage appointments	2. Admin Review	APPOINTMENT; HISTORY	Manage Appointments
Psych update status	Update status	2. Psych Update	APPOINTMENT; SESSION	Psych Dashboard
View history	View history	2. Status display	APPOINTMENT_HISTORY	History / Details
Announcements	Post / View announcements	3. Announcement Flow	ANNOUNCEMENT	Messages & Announcements
Manage records	Manage student records	Admin record access	STUDENT; USERS	Manage Students
Statistics	View statistics	Supporting (data)	COUNSELLING_STATISTICS	Reports & Statistics
Profile	Manage Profile	1. Auth & Profile	USERS; STUDENT	Profile screens